The Coalition/Doomsday Population Model

Purpose

The purpose of this lab is to gain experience in using calculus to model population problems.

Preview

In class you discussed the exponential growth model and the logistic model for population. In this Lab we shall apply some of the techniques which you learned to understand the growth of the world population. In Part I we shall see that neither the exponential nor the logistic model provide good explanations for world population data over the past four centuries. In Part II we introduce the coalition (or doomsday) model, which fits the data well and which predicts rather dire consequences in the future. Finally, in Part III we give data from the United Nations which could affect our view of the coalition model.

Table 1: World Population 1650-1987 (in millions)

t	0	100	150	200	250	270	280	290	300	310	326	337
P	545	728	906	1171	1608	1834	2070	2295	2517	3005	4000	5000

Part I: Previous Population Models

The table above gives the world population in various years between 1650 and 1987. We take the year 1650 to correspond to $t = 0$. Plot this data in a spreadsheet.

1. Investigate to see if this is exponential growth. Recall that if the growth is exponential, then the log of the population should be a linear function of t. Is it? Is the growth faster or slower than exponential?

2. Could this be logistic growth? Recall that in logistic growth the ratio $\frac{1}{P}\frac{dP}{dt}$ decreases as P gets larger. To check this, you will need to approximate $\frac{dP}{dt}$ by using difference quotients.

Part II: The Coalition Model

In your calculations in Part I, you should have found that the ratio $\frac{1}{P}\frac{dP}{dt}$ seems to *increase* as P gets larger. If the ratio increases as a power of P, then

$$\frac{\frac{dP}{dt}}{P} \;=\; kP^r$$

where k and r are some positive constants. This would give us the differential equation

$$\frac{dP}{dt} \;=\; kP^{1+r}$$

Just such a proposal was made by three electrical engineers in 1960 in an article in *Science*. They argued that it is very reasonable for the ratio $\frac{1}{P}\frac{dP}{dt}$ to increase. Unlike simpler species (like bacteria multiplying in a dish), human beings cooperate with each other in utilizing resources. In ancient times, humans cooperated to hunt. In modern times, humans cooperate to invent new technology for more efficient farming, for eradicating disease, and for more efficient communication. The co-operation makes the species more and more capable of utilizing natural resources for growth and, therefore, the ratio $\frac{1}{P}\frac{dP}{dt}$ should increase as P gets larger, since if P is larger there are more people to cooperate. Thus, this differential equation is sometimes called the "coalition model" because individuals act in coalition with others for the betterment of all. This seems like a reasonable idea, though, of course, one can think of lots of examples where human beings don't cooperate well. In this part of the Lab, you are asked to investigate this idea to see if it is borne out by the data.

1. The first question is whether kP^r is a good choice for modeling the increasing ratio $\frac{1}{P}\frac{dP}{dt}$. If so, the population $P(t)$ would satisfy the differential equation. Taking the log of both sides we find:

   $$\ln\left(\frac{dP}{dt}\right) \;=\; \ln k \;+\; (1+r)\ln P$$

   That is, $\ln\left(\frac{dP}{dt}\right)$ would be a linear function of $\ln P$. You have already computed approximate values of $\frac{dP}{dt}$ in Part I. Use your calculations to check whether it is reasonable to model $\ln\left(\frac{dP}{dt}\right)$ versus $\ln P$ as a linear function.

2. Use the method of separation of variables to derive the solution

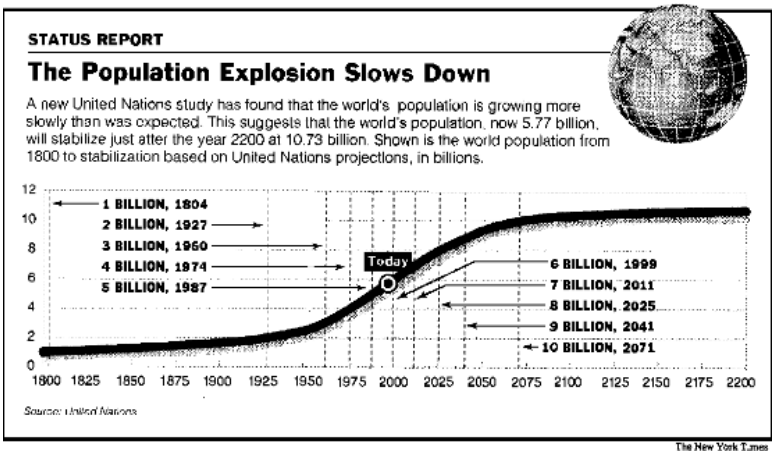
   $$P(t) \;=\; \left(\frac{1}{\frac{1}{P(0)^r} - rkt}\right)^{\frac{1}{r}}$$

   of the differential equation.

3. We know $P(0) = 545$ million people. The constants k and r should be chosen so that the straight line relationship in Question 1 fits the data as best as possible. This not so easy to do and the solution depends sensitively on the choice of k and r. So, assume that we have found reasonable choices $r = .897$ and $k = 10^{-5}$. Using these choices, draw a graph of the solution over a plot of the data in the population table. How well does the coalition model fit the data?

4. Show that there is a year T so that the solution $P(t)$ from Question 2 approaches ∞ as t approaches T (that is, show that the population has a vertical asymptote at $t = T$). How old will you be in year T? The authors of the *Science* article refer to T as "doomsday" for obvious reasons. We note that their doomsday is different from ours because our choices of r and k are somewhat different from their choices.

Part III: Recent Information. In November 17, 1996, the *New York Times* reported that data from the United Nations shows that the "world is less crowded than expected." The article cites data showing that the world population in 1996 was approximately 5.77 billion.



(a) How does the figure 5.77 billion compare with the prediction of the coalition model for the year 1996?

(b) Did the ratio (1) increase or decrease between 1987 and 1996?

References.

1. von Foerster, Heinz, Mora, Patricia M., and Larry W. Amiot, "Doomsday: Friday, 13 November, A.D. 2026," **Science 132**(1960), 1291-1295.

2. Smith, David A., "Human Population Growth: Stability or Explosion?" *Mathematics Magazine* **50** (1970), 186-197.

3. Crossette, Barbara, "The World Is Less Crowded Than Expected, the U.N. Reports," in *The New York Times*, November 17, 1996.